

CENTRIFUGAL COMPRESSOR — OPERATION & MAINTENANCE MANUAL

Equipment Tag: Compressor-C1 | Model: CX-400 Series | Rev. 4 (2022)

IndustraTech Industries Ltd. | Document No: ITI-OM-CX400-001 | CONFIDENTIAL

SECTION 1 — EQUIPMENT OVERVIEW

This manual governs operation, preventive maintenance, and safe shutdown of the Compressor-C1 centrifugal compressor unit installed in Process Area B, Bay 3 of the facility. It supersedes all previous revisions. All personnel performing maintenance must read and acknowledge this document before commencing work on Compressor-C1.

Equipment Tag: Compressor-C1 | Type: Single-stage centrifugal compressor | Model: CX-400 Series | Serial No: CC-2019-00147 | Manufacturer: IndustraTech Industries Ltd. | Installation Date: 2019-03-15 | Design Life: 20 years

SECTION 2 — TECHNICAL SPECIFICATIONS

The following parameters define the design limits and normal operating envelope for Compressor-C1. Do NOT operate outside these limits without written authorisation from the Plant Engineering Department.

Rated Speed: 3,550 RPM

Suction Pressure (normal): 10 – 25 psi

Discharge Pressure (normal): 120 – 145 psi

Maximum Operating Pressure: 150 psi

Design Pressure: 165 psi

Normal Operating Temperature: 180 °F to 220 °F

Maximum Outlet Temperature: 280 °F

Motor Power Rating: 250 kW

Shaft Seal Type: Mechanical (Type III API 682)

Impeller Diameter: 14 inches | Material: 316 Stainless Steel

Bearing Type: Hydrodynamic oil-film bearings, ISO VG 46 lubricant

[WARNING] The Maximum Operating Pressure for Compressor-C1 is 150 psi. This value is determined by the OEM based on the structural limits of the impeller, volute casing, and shaft seals. Sustained operation above 150 psi risks catastrophic seal failure and potential process gas release. This limit may NOT be revised without a formal Management of Change (MOC) review and OEM written concurrence.

SECTION 3 — PREVENTIVE MAINTENANCE SCHEDULE

3.1 Daily Operator Checks

- Verify suction and discharge pressures within normal operating range.
- Check bearing oil level and bearing temperature indicators (normal: < 200 °F).
- Listen for abnormal noise or vibration (baseline: < 3.5 mm/s RMS).
- Record all readings in the Daily Operations Log (Form OPS-DL-001).

3.2 Semi-Annual Inspection (6-Month Interval — Certified Technician Required)

- Disassemble and inspect impeller for erosion, fouling, and cavitation pitting.
- Replace mechanical shaft seal if wear exceeds 0.5 mm on any sealing face.
- Flush and replace bearing lubrication oil (ISO VG 46); inspect oil for metallic particles.
- Inspect and clean suction filter elements; replace if differential pressure > 2 psi.
- Verify shaft alignment to within 0.002 in TIR using laser alignment tool.
- Complete PM checklist Form EQ-PM-009 and submit to Maintenance Supervisor.

3.3 Annual Overhaul

- Full disassembly and dimensional inspection of all rotating parts.
- Non-destructive testing (NDT) of impeller and pressure casing.
- Replace all seals, gaskets, wear rings, and O-rings regardless of condition.
- Alignment certification required from a certified vibration analyst.
- Pressure hydrostatic test to $1.5 \times$ design pressure (247.5 psi) after reassembly.

SECTION 4 — OPERATING PROCEDURES

4.1 Normal Start Sequence

1. Confirm lube oil system is pressurised to minimum 18 psi supply pressure.
2. Open suction isolation valve V-101 slowly over 30 seconds to full open.
3. Start drive motor per Electrical SOP SOP-ELEC-003.
4. Monitor startup current — must return to nameplate amps within 15 seconds.
5. Verify discharge pressure builds to 120–145 psi within 60 seconds of start.
6. Log startup time and initial readings in the Daily Operations Log.

4.2 Normal Shutdown

1. Gradually reduce process load over 5 minutes minimum.
2. Close discharge isolation valve V-102.
3. De-energise motor at MCC Panel B-12.
4. Allow unit to coast to rest — do NOT apply mechanical braking.

5. Close suction isolation valve V-101 after unit reaches complete stop.
6. Log shutdown time and reason in the operations log.

4.3 Emergency Shutdown (ESD)

[WARNING] If discharge pressure exceeds 150 psi, bearing temperature exceeds 250 °F, excessive vibration is detected, or abnormal noise occurs: press ESD pushbutton ESD-C1 immediately. Notify control room and shift supervisor. Do NOT attempt restart without written Engineering clearance.